

1. A CPU has a 16-bit instruction format using an expanding opcode architecture. It supports 3-address instructions and 2-address instructions. Each address field requires 4 bits. If there are K 3-address instructions, what is the maximum number of 2-address instructions that can be formulated?

- (A) $(16 - K) \times 16$ (B) $(16 - K) \times 4$
(C) $16^2 - K$ (D) $2^4 \times (16 - K)^2$

2. A machine uses 32-bit instructions and word-addressable memory. An instruction JMP Offset uses PC-relative addressing. If the PC holds 0x00001000 when fetching this 1-word instruction and the target address is 0x00000F50, what 16-bit signed offset value (in decimal) must be embedded in the instruction?

- (A) -177 (B) -176
(C) -175 (D) -174

3. Consider a 4-stage CPU pipeline with stage delays: $S_1 = 80$ ps, $S_2 = 120$ ps, $S_3 = 90$ ps, $S_4 = 100$ ps. The interface registers introduce a delay of 10 ps each. What is the clock cycle time and total execution time for 100 instructions?

- (A) 130 ps, 13.39 ns (B) 120 ps, 12.36 ns
(C) 130 ps, 13.00 ns (D) 140 ps, 14.42 ns

4. A general register organization contains 32 general-purpose registers, each 32 bits wide. The registers are connected to a common 32-bit bus system via multiplexers. What is the total number of multiplexers required and the multiplexer configuration?

- (A) 32 MUX of size 32:1 (B) 16 MUX of size 32:1
(C) 32 MUX of size 16:1 (D) 64 MUX of size 16:1

5. Which sequence of register transfers correctly represents the execution phase of an indirect memory-reference fetch cycle in a basic computer system?

- (A) $MAR \leftarrow IR(\text{Address}); MBR \leftarrow M[MAR]; IR(\text{Address}) \leftarrow MBR$ (B) $MAR \leftarrow PC; MBR \leftarrow M[MAR]; PC \leftarrow PC + 1$
(C) $IR \leftarrow MBR; MAR \leftarrow PC; MBR \leftarrow M[MAR]$ (D) $MAR \leftarrow MBR; MBR \leftarrow PC; IR \leftarrow M[MAR]$

6. In a displacement addressing mode with index scaling, the Base Register contains 0x2000, the Index Register contains 0x0010, the Scale Factor is 4, and the Displacement offset is 0x0050. What is the Effective Address (EA) in hexadecimal?

- (A) 0x2090 (B) 0x2040
(C) 0x2060 (D) 0x2080

7. A computer uses a 32-bit fixed-length instruction format. The CPU has 64 general-purpose registers, supports 128 opcodes, and directly addresses a 16 MB byte-addressable memory space. What is the total length required for a Register-to-Memory instruction?

- (A) 37 bits (Requires 2-word instruction format) (B) 32 bits (1 word)
(C) 27 bits (5 unused bits) (D) 30 bits (2 unused bits)

8. A 32-bit computer architecture uses 2-level indirect addressing. An instruction specifies memory address $A_1 = 5000$. Memory location 5000 holds value 5020, location 5020 holds value 8000, and location 8000 holds operand 9999. What is the Effective Address (EA) and value loaded into CPU accumulator?

- (A) EA = 5020, Value = 8000 (B) EA = 8000, Value = 9999
(C) EA = 5000, Value = 5020 (D) EA = 9999, Value = 8000

10. A CPU control unit generates 48 distinct control signals. If grouped into 6 mutually exclusive sets of equal size for vertical microinstruction encoding, how many control field bits are saved compared to horizontal microinstruction encoding?

- (A) 30 bits
- (B) 24 bits
- (C) 18 bits
- (D) 36 bits

11. For a zero-operand stack CPU architecture, evaluate the maximum stack depth required to execute the arithmetic expression: $X = (A + B * C) / (D - E * F)$.

- (A) 4 operands
- (B) 5 operands
- (C) 3 operands
- (D) 6 operands

12. What is the theoretical speedup achieved by an 8-stage pipeline CPU over a non-pipelined CPU for a task of 1000 instructions, assuming clock cycle time $t_{\text{pipe}} = 2 \text{ ns}$ and non-pipelined execution time $t_{\text{nonpipe}} = 16 \text{ ns}$?

- (A) 7.94
- (B) 8.00
- (C) 7.42
- (D) 6.85

13. Consider the RTL conditional statement: $P: R2 \leftarrow R1, R1 \leftarrow R2$. What physical hardware event occurs during a clock pulse when control variable $P = 1$?

- (A) Values of $R1$ and $R2$ are swapped simultaneously via edge-triggered registers
- (B) $R2$ receives $R1$ first, then $R1$ receives its original value back
- (C) A hardware bus conflict occurs causing undefined states
- (D) Both registers are cleared to zero

14. In an isolated I/O scheme compared to a memory-mapped I/O scheme, how are input/output operations distinguished from main memory access operations?

- (A) Distinct IN and OUT machine instructions with separate I/O read/write control lines
- (B) By using high-order memory address bits as I/O flags
- (C) By running I/O operations in kernel mode only
- (D) By using an external direct memory access controller exclusively

15. During the interrupt cycle of a CPU, which sequence of micro-operations correctly saves the return address and branches to the Interrupt Service Routine (ISR) at vector location $0x0008$?

- (A) $MBR \leftarrow PC; MAR \leftarrow SP; M[MAR] \leftarrow MBR; PC \leftarrow 0x0008$
- (B) $MAR \leftarrow PC; MBR \leftarrow SP; PC \leftarrow 0x0008$
- (C) $SP \leftarrow SP + 1; M[SP] \leftarrow PC; PC \leftarrow 0x0008$
- (D) Both (A) and (C) depending on stack growth direction

16. In a hardwired control unit, if control signal $Z_{in} = T_1 \cdot (ADD + SUB) + T_2 \cdot LDA$, how many AND gates and OR gates at minimum are required to construct this control logic function?

- (A) 2 AND, 2 OR gates
- (B) 3 AND, 1 OR gates
- (C) 2 AND, 1 OR gates
- (D) 4 AND, 2 OR gates

17. In Auto-decrement Addressing mode used with a stack pointer (SP), when does the decrement operation occur relative to the memory access?

- (A) Before the memory location is accessed (Pre-decrement)
- (B) After the memory location is accessed (Post-decrement)
- (C) Simultaneously with memory access via dual bus
- (D) Controlled by CPU flag register after instruction commit

18. Which addressing mode allows an instruction to specify a memory operand location relative to the current instruction address plus a constant offset, ensuring position-independent code execution?

- (A) PC-Relative Addressing Mode
- (B) Indexed Addressing Mode

19. In Base-Register addressing mode, a program is loaded starting at physical segment address 0x00040000. If the logical relative address inside the instruction is 0x00002450, what is the physical Effective Address (EA)?

- (A) 0x00042450 (B) 0x0003DBB0
(C) 0x00024500 (D) 0x00040000

20. A CPU executes a program in 4.0 seconds. The CPU clock frequency is 2.0 GHz (2×10^9 cycles/sec). If the program contains 5×10^9 instructions, what is the average CPI (Cycles Per Instruction)?

- (A) 1.60 (B) 1.25
(C) 2.00 (D) 2.50

21. A 32-bit variable-length instruction set architecture uses 1-byte, 2-byte, and 4-byte instruction formats. If the Memory Data Bus is 32 bits wide and aligned on 4-byte boundaries, how many memory fetch cycles are required to fetch an unaligned 4-byte instruction spanning two 32-bit memory words?

- (A) 2 memory cycles (B) 1 memory cycle
(C) 4 memory cycles (D) 3 memory cycles

22. A 32-bit CPU has a clock rate of 2.5 GHz. If a benchmark program requires 10^9 instructions and runs with an average CPI of 1.25, what is the total CPU execution time?

- (A) 0.50 seconds (B) 0.25 seconds
(C) 0.75 seconds (D) 1.00 seconds

23. In a microprogrammed control unit, if an instruction opcode is 6 bits wide and mapped to a 12-bit control memory address by appending zeros to the right, what is the starting microinstruction address for opcode 011010 in binary?

- (A) 011010000000 (B) 000000011010
(C) 011010000000 (D) 110100000000

24. A 5-stage CPU pipeline encounters a conditional branch instruction at stage 2 (Decode). If the branch outcome is determined at stage 4 (Execute) and the branch is taken, how many clock cycles are lost as pipeline penalty stall cycles?

- (A) 2 clock cycles (B) 3 clock cycles
(C) 1 clock cycle (D) 4 clock cycles

25. A RISC processor uses overlapping register windows with 8 global registers, 8 local registers, 8 in registers, and 8 out registers per window. For a system with 4 register windows, what is the total number of physical general-purpose registers in the register file?

- (A) 72 registers (B) 104 registers
(C) 128 registers (D) 64 registers

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(A)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(B)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(D)	20	(A)	25	(A)